

# AY25 PROGRAM ASSESSMENT REPORT

v2026.1

## Tracking data

|                                                                                  |                                                                                                                       |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <a href="#">CIM code</a>                                                         | PHD-PHYS-4                                                                                                            |
| Program name                                                                     | PhD in Physics                                                                                                        |
| Assessment Year                                                                  | 2024/2025                                                                                                             |
| Program assessment coordinators                                                  | Grant Bunker / Pavel Snopok                                                                                           |
| Year of Current Assessment Plan                                                  | AY24                                                                                                                  |
| No assessment this year <b>per plan</b>                                          | <input type="checkbox"/>                                                                                              |
| LO(s) assessed                                                                   | LO#3 and LO#4                                                                                                         |
| Presented at program authority meeting date:<br><br><i>Please attach minutes</i> | Report was distributed to departmental leadership, assessment committee, and relevant instructors prior to submission |

## Methodology

According to the program assessment plan, in academic-year 2024/2025, the following learning objectives were assessed:

3. Demonstrate understanding of **quantum mechanics** by being able to solve problems (at the graduate level)
4. Demonstrate understanding of **thermal and statistical physics** by being able to solve problems (at the graduate level)

## Methodology for Assessment of Learning Objective #3 – Quantum Mechanics

For the purposes of assessment of LO#3, artifacts were collected from PHYS-510: Quantum Theory II. The artifacts were student submissions of the in-class final exam.

This course contained 9 students from several different programs. Each student on the roster was assigned an identifier (i.e. student 1 through student 9), but only students who are enrolled in this program are listed in the results sections below, which is why the numbers are not sequential. Also note that the rosters for PHYS-510 and PHYS-515 are not identical, so student #1 in the data table for LO#3 is not necessarily the same as student #1 in the data table for LO#4.

The problem statement is below:

---

**Consider a three-dimensional repulsive ( $V_0 > 0$ ) square barrier of radius  $a$ .**

$$V(r) = \begin{cases} V_0 & (r < a) \\ 0 & (r \geq a) \end{cases}$$

**A particle of energy  $E = \frac{\hbar^2 k^2}{2m} < V_0$  is incident on the well. Calculate the scattering amplitude and differential cross-section in the Born approximation for this potential.**

---

This is a classic quantum mechanical scattering problem. The problem asks students to demonstrate an understanding of plane wave solutions to the Schrodinger equation, differential cross-sections, and using the Born approximation to calculate scattering amplitudes. Because scattering is ubiquitous in many fields of physics (especially particle and nuclear physics), this problem is well-aligned with learning objective #3 of this program.

The rubric used to assess these problems is given below

| Problem part, or skill assessed | Weight | DOES NOT MEET EXPECTATIONS<br>1 - 2                                                                                                                   | MEETS WITH CONCERNS<br>3 - 4                                                                                                                                                                                                | MEETS EXPECTATIONS<br>5 - 7                                                                                                                                                                                      | EXCEEDS EXPECTATIONS<br>8                                                                                                                                                                         |
|---------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROBLEM 2                       | 1      | does not successfully construct a plausible and justified expression for differential scattering cross section in Born approximation for hard sphere. | By applying the Born approximation, only partially succeeds in Fourier transforming the given potential to obtain a correct expression for the differential scattering cross section, with significant errors or omissions. | By applying the Born approximation, successfully Fourier transforms the given potential to obtain a mostly correct expression for the differential scattering cross section with only minor errors or omissions. | By applying the Born approximation, successfully Fourier transforms the given potential to obtain a correct expression for the differential scattering cross section with no errors or omissions. |

## Methodology for Assessment of Learning Objective #4 – Thermal and Statistical Physics

For the purposes of assessment of LO#4, artifacts were collected from PHYS-515 –Statistical Mechanics. The artifacts were student final exams.

This course contained 6 students. Each student on the roster was assigned an identifier (i.e. student 1 through student 1), but only students who are enrolled in this program are listed in the results sections below, which is why the numbers are not sequential.

The following problem was used for assessment

---

**The pressure  $P$  of a gas is related to its density  $n = N/V$  and temperature  $T$  by the truncated expression**

$$P = k_B T n - \frac{b}{2} n^2 + \frac{c}{6} n^3$$

**Where  $b$ , and  $c$  are assumed to be positive temperature-independent constants.**

- (a) Locate the critical temperature  $T_c$  below which the equation must be invalid and the corresponding density  $n_c$  and the pressure  $P_c$  of the critical point. Find the ratio  $\frac{k_B T n_c}{P_c}$ .**
- (b) Calculate the isothermal compressibility  $\kappa_T = -\frac{1}{V} \left( \frac{\partial V}{\partial P} \right)_T$  and sketch its behavior as a function of  $T$  for  $n = n_c$**
- (c) On the critical isotherm, give an expression for  $(P - P_c)$  as a function of  $(n - n_c)$ .**

---

This problem assesses whether students are comfortable with thermodynamic relations between pressure, volume, density, and temperature, and their understanding of critical point phenomena. Mastery of problems on these topics are central to this program's learning goals.

The rubric used to assess this problem is given below.

| Problem or part | Weight | DOES NOT MEET EXPECTATIONS<br>1 - 2                                                                                                                                                                                               | MEETS WITH CONCERNS<br>3 - 4                                                                                                                                                                                                                                                           | MEETS EXPECTATIONS<br>5 - 7                                                                                                                                                                                                                                              | EXCEEDS EXPECTATIONS<br>8                                                                                                                                                                                                                                                  |
|-----------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROBLEM 2a      | 0.43   | Does not recognize required conditions for critical point, or hows significant gaps in conceptual understanding or mathematical application, or does not engage with the specific given equation of state, or does not show work. | From the given equation of state, recognizes the critical point is determined by the condition that the $P'(n)=0$ and $P''(n)=0$ . From that, the critical temperature and pressure and their ratio are correctly determined, but with significant conceptual or computational errors. | From the given equation of state, recognizes the critical point is determined by the condition that the $P'(n)=0$ and $P''(n)=0$ . From that, the critical temperature and pressure and their ratio are incorrectly determined. but with minor errors, not conceptual in | From the given equation of state, recognizes the critical point is determined by the condition that the $P'(n)=0$ and $P''(n)=0$ . From that, the critical density $n_c$ , temperature $T_c$ , and pressure/ $P_c$ and the ratio $T_c/(P_c n_c)$ are correctly determined. |
| PROBLEM 2b      | 0.29   | did not successfully calculate a plausible expressions without significant omissions or mathematical or conceptual errors, or does not show work.                                                                                 | using results from part a, produce definition of isothermal compressibility, calculate it for the given equation of state, and sketch its divergence as $T$ approaches $T_c$ . Some significant conceptual or mathematical errors.                                                     | using results from part a, produce definition of isothermal compressibility, calculate it for the given equation of state, and sketch its divergence as $T$ approaches $T_c$ . Only minor conceptual or mathematical errors. Or omits sketch.                            | using results from part a, produce definition of isothermal compressibility, calculates it for the given equation of state, and sketch its divergence as $T$ approaches $T_c$ . No significant conceptual or mathematical errors.                                          |
| PROBLEM 2c      | 0.29   | did not successfully calculate a plausible expressions without significant omissions or mathematical or conceptual errors, or does not show work                                                                                  | Using results from parts a and b, student finds incorrect expression for $P$ - $P_c$ vs $(n-n_c)$ on the critical isotherm. Some significant conceptual or mathematical errors.                                                                                                        | Using results from parts a and b, student successfully finds correct or near-correct expression for $P$ - $P_c$ vs $(n-n_c)$ on the critical isotherm. Only minor conceptual or mathematical errors.                                                                     | Using results from parts a and b, student successfully finds correct expression for $P$ - $P_c$ vs $(n-n_c)$ on the critical isotherm. No significant conceptual or mathematical errors.                                                                                   |

Each artifact was evaluated according to the above rubric(s), and a weighted average was used to arrive at a final score ranging from 0 to 8. A letter grade was assigned for each Learning Objective as follows:

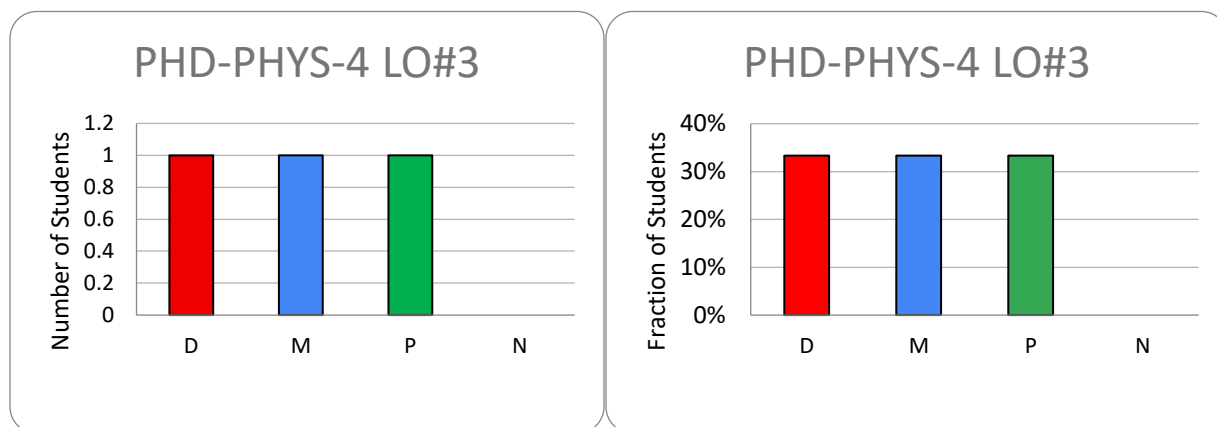
| Designation                    | Score Range |
|--------------------------------|-------------|
| D (Does Not Meet Expectations) | 0 - 4       |
| M (Meets Expectations)         | 4 - 6.5     |
| P (Proficient)                 | 6.5 – 8.0   |

\*Note that a student receives N (Not Assessed) in cases where the student withdrew from the course, took an incomplete, or when an artifact is not available due to issues outside of the student's control. A student who is non-compliant (i.e. does not submit an assigned assessment) receives D (Does not Meet Expectations).

## Results for LO#3 – Quantum Mechanics

The classlists for PHYS-510 and PHYS-515 were not identical; student #1 in the LO#3 data table is not necessarily the same as student #1 in the data table for LO#4

| student   | major    | D/M/P |
|-----------|----------|-------|
| student 2 | PHYS-PHD | P     |
| student 6 | PHYS-PHD | D     |
| student 7 | PHYS-PHD | M     |



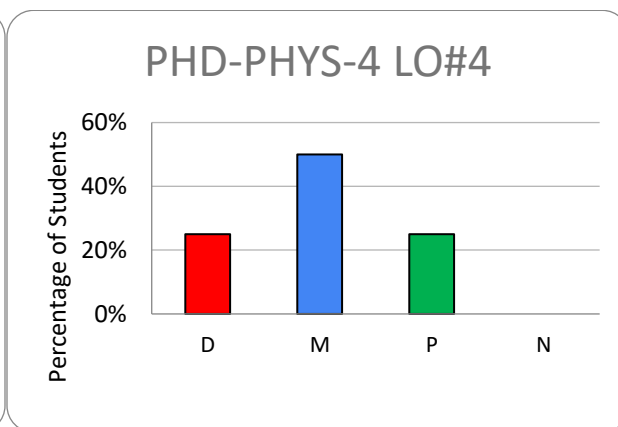
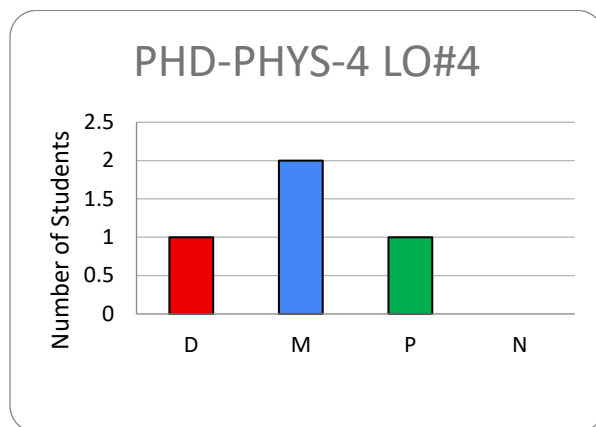
## Artifact samples for LO#3 – Quantum Mechanics

The following sample artifacts are included with this report

- Student 2 - Assessed as Proficient (P)
- Student 6 - Assessed as Meets Expectations (M)
- Student 7 - Assessed as Does Not Meet Expectations (D)

## Results for LO#4 – Thermal and Statistical Physics

| student   | major    | D/M/P |
|-----------|----------|-------|
| student 3 | PHYS-PHD | M     |
| student 4 | PHYS-PHD | P     |
| student 5 | PHYS-PHD | M     |
| student 6 | PHYS-PHD | D     |



## Artifact samples for LO#4 – Thermal and Statistical Physics

Two example artifacts are included with this report

- Student 4 – Assessed as Proficient (P)
- Student 5 – Assessed as Meets Expectations (M)
- Student 6 – Assessed as Does not Meet Expectations (D)

## Description of findings

### LO#3

Approximately 66% of students (2/3) were assessed as meeting the learning objectives of the program (grades of M or P). This is a small data set, so despite it appearing as though 33% of students did not meet the LO, this corresponds to only a single student.

There is only one student who was assessed as (D). It is worth noting that the exam from Student #6 who was assessed as (D) – does not show an attempt at problem #2 (the problem used for assessment of this learning objective). It is, therefore, unclear whether this student should instead be assessed as (N) – not assessed. The lack of an attempt at the problem could be an indication that the student lacks the skills and knowledge to complete it, but it could also be an indication that the student ran out of time, or that there were issues with the scanning of the exam. It is difficult to draw conclusions as to the strength of the student's understanding (or lack thereof) based on the present data.

We therefore maintain that the data support the conclusion that most (if not all) students are meeting the learning objectives, and we cannot recommend any concrete continuous improvement plan based on the present data.

### LO#4

Approximately 75% of students (3/4) were assessed as meeting the learning objectives of the program (grades of M or P). This is a small data set, so despite it appearing as though 25% of students did not meet the LO, this corresponds to only a single student.

It is worth noting that the student in question (student #6 in PHYS-515), was also the only student assessed as (D) in last year's report for LO#2 (student #7 in PHYS-505 in Fall 2023). These two learning objectives (LO#2 – Classical Mechanics and LO#4 – Thermal/Statistical Physics) are largely independent of each other, these two courses do not directly build on each other, and PHYS-505 is not a prerequisite for PHYS-515.

Because this student struggled to meet expectations for two learning objectives across years, it is tempting to speculate that the student's undergraduate preparation may have been weak, leading to a lack of foundational knowledge or mathematical skills. This lack of preparation could lead to difficulties across the curriculum. It would be interesting to investigate which knowledge or skills the student may be lacking from their undergraduate degree, and whether initiatives could be employed to fill those gaps. However, it is unwise to over-react based on one data point. We caution that any investigations or the implementation of any continuous improvement initiatives and investigations would require further evidence of a broad lack of preparation beyond this single student.

## Continuous Improvement Plans

### Prior CI plan impact

☒ no previous CI plans available

This cycle's CI plan

☒ After consideration, no CI plans are recommended.